

PIPE-Cypher: Automatic Enterprise Benchmark Generation for Text-to-Cypher Systems

Anonymous ACL submission

Abstract

Enterprises increasingly use property graphs and Cypher to analyze fraud, risk, identity, customer, and operational data. Public Text2Cypher benchmarks rarely reflect private schemas, terminology, access patterns, or difficulty distributions. We present PIPE-Cypher, a local-model pipeline for generating organization-specific natural-language-to-Cypher benchmarks. PIPE-Cypher combines schema introspection, reverse-query grounding, constrained Cypher generation, deterministic read-only and schema validation, execution feedback, repair, diversity controls, and LLM-judge review. The system is designed for industry settings where benchmark generation must avoid paid APIs, preserve data governance, and remain repeatable as graphs evolve.

1 Introduction

Property graphs encode enterprise relationships directly: account transfers, identity entitlements, access paths, customer interactions, and fraud rings. Cypher is a common query language for these graphs. As LLMs become natural-language interfaces for graph analytics, organizations need reliable benchmarks for natural-language-to-Cypher systems on their own schemas.

Static public benchmarks are insufficient for this setting. They cannot contain private labels, relationship types, categorical values, or domain-specific wording, and they do not track schema evolution. PIPE-Cypher addresses this gap by generating private, repeatable, execution-grounded NL-Cypher benchmarks.

2 Related Work

Recent Text2Cypher resources, including Mind the Query (Chauhan et al., 2025), SyntheT2C (Zhong et al., 2025), Auto-Cypher (Tiwari et al., 2025), and Text2Cypher (Ozsoy et al., 2025), show that

high-quality Cypher data requires more than asking an LLM for query pairs. These works motivate schema grounding, execution checks, synthetic generation, verification, and complexity-aware evaluation. PIPE-Cypher differs by targeting private enterprise benchmark generation as the primary artifact: organizations bring their own graph, run local models, enforce Cypher constraints, and receive an executable benchmark with diversity and judge metadata.

Text-to-SQL benchmarks such as Spider 2.0 (Lei et al., 2024) and BIRD (Li et al., 2023) motivate realistic database tasks, execution-based evaluation, and enterprise workflow complexity. CIKM AutoQuery (Zheng et al., 2024) motivates strategy-level workload generation analysis and separating workload quality from model quality. LDBC FinBench (Qi et al., 2023) and SNB (Püroja et al., 2023) provide graph workloads suitable for industry-oriented evaluation.

For generated benchmark quality, PIPE-Cypher combines text-generation diversity diagnostics with text-to-query structure metrics. Distinct-n (Li et al., 2016) and self-BLEU-style redundancy checks (Zhu et al., 2018) capture lexical repetition, while schema coverage, query-signature diversity, structural feature rates, and normalized entropy over graph/category/difficulty cells capture properties that matter for executable Cypher benchmarks.

3 Method

PIPE-Cypher has six stages: schema profiling, template generation, reverse Cypher grounding, constrained Cypher generation and repair, deterministic validation and execution, and LLM-judge review. Deterministic checks enforce read-only safety, syntax shape, schema-visible labels and properties, explicit relationship types, observed relationship directions, schema-provided categorical values, and non-empty execution where required. A

lightweight Cypher analyzer extracts return aliases, variables, labels, relationship observations, risky constructs, and rewrite skip reasons so normalization is auditable rather than a silent string edit. The direction gate interprets both outgoing and incoming Cypher arrow syntax before schema checking, and rejects undirected relationship patterns so directionality is an executable constraint rather than only a prompt instruction.

4 Implementation

The implementation is a Python package with Neo4j as the experimental backend. The paper frames the contribution around Cypher and property graphs rather than a single vendor. Local models are served through a vLLM/OpenAI-compatible endpoint on ds-serv6, using Qwen3.5-9B for generation and judging.

All reported generation, judging, and downstream evaluation runs use a local Qwen3.5-9B endpoint. This keeps the study aligned with an enterprise deployment pattern in which the benchmark pipeline runs inside the organization’s compute boundary without paid generation APIs.

The built-in FinBench profile is grounded in the public datagen snapshot export and includes typed node and relationship properties. Live schema inspection also records low-cardinality string values as categorical constraints for prompting and validation. The generated Neo4j import script uses uniqueness constraints for nodes and creates, rather than merges, transaction relationships so repeated account-to-account events are preserved. The SNB reference profile is grounded in the official Neo4j/Cypher headers and read-query files.

For generation, PIPE-Cypher uses seeded workload templates with deterministic reverse-binding queries and a mixed mode that prepends proven workload seeds to LLM-proposed templates. Every run records accepted and rejected candidates for yield analysis. Retrieved few-shot examples replace graph-specific values with typed placeholders, preserving query structure while reducing tenant-value leakage and memorized entity reuse. A dependency-light value grounder adds typed prompt annotations for categorical values and reverse-bound entities, covering punctuation variants, possessives, plurals, synonyms, name partials, and small typos.

Inspired by Mind the Query’s prompt-setting analysis, the implementation exposes prompt pro-

files for schema-only, instructions-only, examples-only, examples-plus-instructions, and full governed PIPE-Cypher generation. These profiles are configured as ablation variants and must pass the same target-size and paper-readiness standards before any result is reported.

For LLM-judge review, the implementation slices schema context to the labels, relationship types, and properties used by the candidate query. This preserves validation against the full schema while keeping local 9B prompts within context limits on larger schemas.

The deterministic Cypher layer uses production-derived constraints: schema-only prompting, exact matching, relationship direction discipline, RETURN DISTINCT, reserved variable rejection, categorical values, required contextual return columns, fuzzy value annotations, placeholderized retrieval examples, and parser-aware rewrite boundaries. PIPE-Cypher records parser-style structure features and skips rewrites for risky constructs such as UNION, CALL, UNWIND, WHERE EXISTS, multiple WHERE clauses, or reserved variables.

We also estimate seed-template capacity before full generation. This check caught and fixed a scale blocker: reverse-binding execution was hard-capped to 10 rows, and no-slot negation/ranking seeds could not support full category targets. PIPE-Cypher now uses the configured binding limit and includes slotted negation and ranking seeds for both graphs.

For arbitrary enterprise onboarding, PIPE-Cypher includes a deployment template for a company’s own graph, read-only credentials, local model endpoint, schema introspection, privacy policy, dry run, scaled run, audit, and redacted export. Configurable value-sampling policies bound which low-cardinality graph values enter schema prompts, and redacted exports replace quoted literals, entity values, and string-valued result samples with stable placeholders for broader internal review. The ICIJ onboarding run further motivated schema-derived sparse-category templates: PIPE-Cypher now derives relationship-count, anti-join, and top-k templates from observed labels, relationship directions, and safe low-cardinality properties, then grounds slot values with outcome-aware reverse Cypher.

5 Experiments

The generated benchmark contains 3,000 accepted examples: 2,000 from LDBC FinBench and 1,000

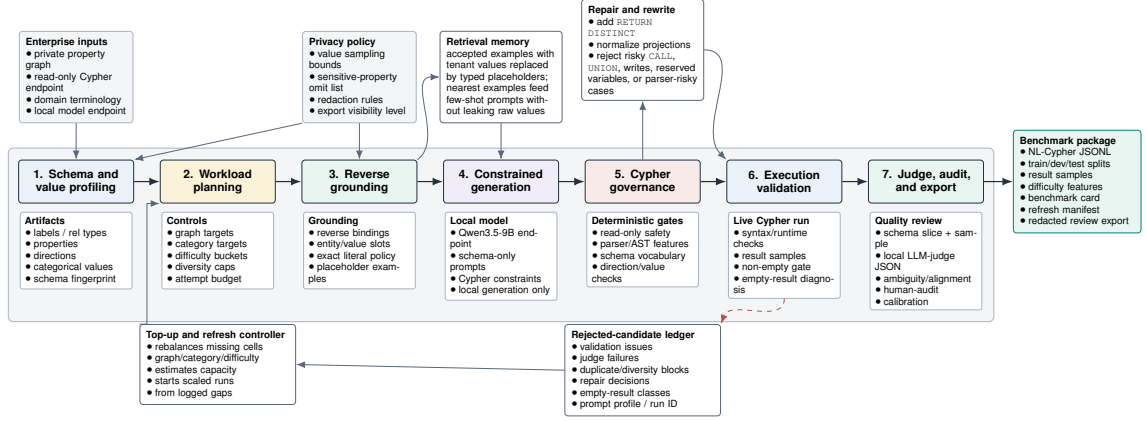


Figure 1: PIPE-Cypher benchmark generation pipeline. The key industry additions beyond static Text2Cypher dataset construction are privacy/value policies, reverse grounding, Cypher governance, execution diagnostics, local judge calibration, and benchmark export/refresh.

Gate	Evidence checked	Failure mode caught
Read-only safety	blocked Cypher write/admin tokens	destructive or operational queries
Schema validity	labels, relationship types, properties, categorical values	hallucinated graph vocabulary or values
Direction validity	observed relationship directions	reversed or invalid traversals
Cypher analysis and normalization	structure extraction, rewrite safety, RETURN DISTINCT	unsafe rewrites or duplicate/noisy rows
Question constraints	quoted values use exact literals	fuzzy matching of exact user values
Execution	query runs and returns rows	syntactic validity without answerability
LLM judge	ambiguity, semantic alignment, schema use	executable but semantically weak examples

Table 1: PIPE-Cypher candidate acceptance gates.

Graph	Target/category	Binding limit	Min seed capacity	All categories meet target
FinBench	250	300	300	Yes
SNB	125	200	200	Yes

Table 2: Built-in seed-template capacity after removing the reverse-binding 10-row cap and adding slotted negation/ranking seeds. Capacity is a theoretical scale-readiness check; final examples still must pass validation, execution, diversity, and judge gates.

from LDBC SNB. Categories include simple retrieval, complex retrieval, simple aggregation, complex aggregation, boolean existence, negation/difference, path/temporal transaction, and ranking/top-k. Appendix Table 8 reports graph statistics, including the audited ICIJ third-graph onboarding workload.

We report three completed ablation suites over FinBench and SNB: an initial target-50 suite, a corrected target-100 suite, and a seed-17 target-50 repeat. Each suite contains 14 graph/setting cells over unconstrained local generation, reverse-only generation, validators+repair, no retrieval, no rewrite, no LLM judge, and full PIPE-Cypher. All paper-reported ablations have collection manifests, model IDs, code revisions, run summaries, and readiness

audits. We additionally report a live ICIJ Offshore Leaks onboarding run as public finance/compliance evidence for arbitrary-schema deployment beyond LDBC.

Downstream Text2Cypher evaluation uses execution accuracy and answer-set precision/recall/F1 as primary correctness metrics. The benchmark evaluator also supports supplementary reference-based text metrics over serialized answer sets and query strings, including ROUGE, BLEU, METEOR, BERTScore, FrugalScore, cosine similarity, Jaro-Winkler similarity, and exact match; Appendix Table 22 lists the supported metrics and citations. We treat these as debugging and near-match diagnostics rather than substitutes for executable correctness.

6 Results

The full live run produced 3,000 accepted examples from 4,777 candidates using local Qwen3.5-9B for generation and judging. Category-specific recovery top-ups filled the only under-target categories from the initial sequential run. Every exported example passed read-only, syntax, schema, execution, non-

Setting	Value
Accepted examples	3,000
Primary graph	LDBC FinBench, 2,000 examples
Secondary graph	LDBC SNB, 1,000 examples
Categories	8 balanced categories, 375 each
Generation/judge model	Local Qwen3.5-9B
Execution backend	Neo4j Community, two live databases
Judge audit packet	80 sampled accepted/rejected pairs

Table 3: Full live experimental setup used for the generated benchmark artifact.

Graph	Candidates	Accepted	Acceptance	Categories at target
FinBench	3,404	2,000	0.588	8/8
SNB	1,373	1,000	0.728	8/8
Total	4,777	3,000	0.628	16/16

Table 4: Full live generation with local Qwen3.5-9B. Candidate counts include the initial sequential run and category-specific recovery top-ups.

empty result, and judge gates.

The accepted full-run records were exported into a benchmark package with stable example identifiers, train/dev/test splits, result samples, gate metadata, aggregate statistics, and a manifest hash for artifact tracking. The export contains 3,000 accepted examples: 2,000 FinBench, 1,000 SNB, and 375 accepted examples in every planned category.

We report diversity as a diagnostic, not only as a design target. The full export has perfect category balance and near-perfect difficulty balance, uses 1,115 unique grounded entity values, and exactly quotes grounded values in 82.6% of examples with entity bindings. The reported local-model run still has low query-signature diversity because seeded graph-grounded templates are doing substantial work. This is useful evidence for industry deployment: the artifact is balanced and executable, while the appendix makes template and value concentration visible rather than hiding it.

The full-run failure taxonomy is concentrated in diversity/duplicate controls during recovery and empty execution results; only 2/4,777 candidates were schema-invalid after deterministic Cypher governance.

For judge calibration, the released artifacts include an 80-row audit packet sampled from accepted and rejected full-run candidates, with a 40/40 judge accept/reject split and coverage over both graphs and all eight categories. A completed human audit shows 80.0% agreement, Cohen’s $\kappa = 0.60$, judge precision and specificity of 1.00, recall of 0.714, and no false accepts in the sample. The judge is therefore conservative: it protects

Export artifact	Examples	FinBench	SNB	Train	Dev	Test
Live full benchmark	3,000	2,000	1,000	2,408	296	296

Table 5: Accepted live full benchmark package with stable IDs, gate metadata, result samples, statistics, and manifest hash `8bc79a53a06b291a`.

Artifact property	Value
Categories	8 balanced categories
FinBench/category	250
SNB/category	125
Difficulty split	1,521 easy / 1,479 medium
Unique labels / rel. types	7 / 9
Read/syntax/schema/exec/judge gates	3,000/3,000

Table 6: Distribution and gate summary for the exported full benchmark artifact.

accepted-example quality while rejecting some human-approved candidates and reducing yield. Human labels are used only for post-hoc calibration, not as a generation gate.

Finally, we ran a downstream Text2Cypher evaluation using local Qwen3.5-9B on the exported full test split. The model saw schema text and the natural-language question, generated Cypher, and was evaluated by live execution against the corresponding FinBench or SNB database. The result verifies that the exported artifact can drive downstream model evaluation and gives a difficulty signal: the zero-shot 9B baseline solves simple aggregation examples but struggles on more operational categories.

The scaled ablation suites show that the benchmark factory is stable once deterministic grounding and schema metadata are fixed. In the target-100 suite, every non-unconstrained graph/setting cell reached all eight category targets with 800 accepted examples per cell; the full PIPE-Cypher setting accepted 800/824 candidates on both FinBench and SNB. Across the three evidence-ready suites, target-normalized coverage is 1.000 for every non-unconstrained cell, while the unconstrained local generation baseline produced no accepted examples under the same execution-grounded gates. ICIJ onboarding reaches the same per-category target on a larger public finance/compliance graph: 800 accepted examples from 983 candidates over 2.0M nodes and 3.3M relationships, with 100 examples in every category.

7 Industry Use

PIPE-Cypher is intended to be rerun when an enterprise graph changes. The generated artifact records

Split	Examples	Parse	Schema	Exec. success	Exec. acc.	Answer F1
Live full test	296	0.959	0.905	0.622	0.189	0.189

Table 7: Downstream Text2Cypher evaluation for local Qwen3.5-9B on the exported full benchmark test split.

the schema snapshot, graph profile, model identifier, validation gates, execution samples, judge scores, difficulty features, and source run for every example. Long-running jobs can be monitored from JSONL records and recovered with category-specific top-up runs that reject questions accepted in earlier runs. This design supports private benchmark refreshes without exposing schemas or values to paid APIs, while still leaving audit hooks for data owners to sample accepted and rejected examples.

The deployment path is designed for arbitrary enterprise schemas: teams provide read-only graph credentials, introspect labels/relationships/properties, choose a bounded categorical-value sampling policy, connect a local model endpoint, run a dry pass, scale generation, calibrate the judge, and export either raw internal artifacts or redacted review artifacts. This makes the FinBench/SNB study a reproducible evaluation setting rather than a hard-coded graph assumption.

8 Conclusion

PIPE-Cypher provides a practical path for enterprises to create private, executable, balanced NL-to-Cypher benchmarks. The pipeline combines schema introspection, constrained generation, deterministic validation, execution feedback, diversity control, and automated judge review.

Limitations

Execution validity does not guarantee semantic correctness. The completed audit suggests a conservative judge with no false accepts in the labeled sample, but larger audits may reveal additional failure modes. Generated artifacts may contain private schema details or sampled values, so organizations should apply privacy redaction and normal data governance controls before broad sharing.

Ethics Statement

PIPE-Cypher is designed for private benchmark generation under local-model deployment constraints. Organizations using it should restrict benchmark artifacts to authorized users, review sampled values for sensitive content, and docu-

ment whether judge calibration relied on human annotators.

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Graph	Nodes	Relationships	Labels	Rel. types	Paper status
FinBench	10,006	57,622	5	9	reported
SNB	34,735	70,842	14	15	reported
ICIJ Offshore Leaks	2,016,523	3,339,267	5	14	onboarding audit

Table 8: Study graph statistics. ICIJ Offshore Leaks is used as a public third-graph onboarding audit for arbitrary finance/compliance schemas beyond the two LDBC study workloads.

ICIJ onboarding property	Value
Graph nodes / relationships	2,016,523 / 3,339,267
Labels / relationship types	5 / 14
Generated / accepted	983 / 800
Acceptance rate	0.814
Categories at target	8/8
Paper audit	ready
Sparse schema-derived accepts	complex agg. 97, negation 28, ranking 98

Table 9: ICIJ Offshore Leaks third-graph onboarding audit. The public finance/compliance graph tests arbitrary-schema generation beyond the two LDBC study workloads; raw values remain outside the paper artifacts.

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A Additional Results

B Claim–Evidence Map

This section maps the main paper claims to the strongest current evidence and to the remaining risks. This is intended as a reviewer-facing audit surface: claims with pending evidence remain marked as such rather than being folded into the main results.

- Claim 1.** PIPE-Cypher can generate a private, executable NL-to-Cypher benchmark

Metric	Value
Question Distinct-1	0.041
Question Distinct-2	0.088
Question self-BLEU-2 (sampled)	0.826
Unique query-signature ratio	0.010
Top query-signature share	0.083
Category normalized entropy	1.000
Graph-category normalized entropy	0.980
Difficulty normalized entropy	1.000
Label coverage	0.412
Relationship-type coverage	0.375
Property-name coverage	0.407
Unique grounded-value ratio	0.373
Grounded values exactly quoted	0.826
Aggregation / negation / ordering rates	0.500 / 0.125 / 0.125

Table 10: Diversity diagnostics for the full exported benchmark. Distinct-n follows text-generation usage; self-BLEU is lower when questions are less redundant; grounded-value metrics are aggregate-only and do not list raw values.

Failure bucket	Count	Share of rejected
Diversity/duplicate control	1,335	0.751
Empty result	374	0.210
Judge semantic reject	66	0.037
Schema invalid	2	0.001

Table 11: Failure taxonomy over full-run generation candidates before benchmark export. Accepted examples are excluded from the bucket shares.

over enterprise-style property graphs using local models.

Evidence. The full local Qwen3.5-9B run exported 3,000 accepted examples: 2,000 FinBench and 1,000 SNB, with 375 examples in every planned workload category and all accepted examples passing read-only, syntax, schema, execution, non-empty-result, and judge gates.

Key artifacts. benchmark export; manifest snapshot; paper table: benchmark export; paper table: full results

Status. Supported by live local-model evidence.

Risk. Generation quality may change with private enterprise schemas beyond FinBench/SNB.

- Claim 2.** Cypher-specific governance makes generated benchmark candidates safer and more auditable than prompt-only generation.

Evidence. The pipeline validates read-only safety, schema-visible labels, node and relationship properties, relationships, observed

PIPE-Cypher category	Closest MTQ category	Added enterprise distinction
Simple retrieval	SR	Direct node/edge lookup with exact filters.
Complex retrieval	CR	Multi-hop or multi-pattern retrieval.
Simple aggregation	SA	Single aggregation such as count/min/max/average.
Complex aggregation	CA	Grouped or multi-stage aggregation over graph neighborhoods.
Boolean existence	EQ	Precise yes/no or existence answer.
Negation/difference	CR	Absence, anti-join, or difference query.
Path/temporal transaction	CR/CA	Temporal or path-oriented transaction neighborhood.
Ranking/top-k	SA/CA	Ordered top-k query with explicit limit.

Table 12: Category crosswalk to Mind the Query. PIPE-Cypher keeps the familiar retrieval/aggregation/evaluation-query structure while adding enterprise workloads such as negation, temporal paths, and ranking.

Gate or ledger	Count	Denominator
Export accepted	3,000	3,000
Read-only safety	3,000	3,000
Syntax validity	3,000	3,000
Schema/value validity	3,000	3,000
Execution success	3,000	3,000
LLM judge pass	3,000	3,000
Rejected candidates logged	1,777	4,777

Table 13: PIPE-Cypher validation cascade for the full export and its logged candidate ledger. Unlike Mind the Query, human review is calibration-only.

relationship direction, categorical values, contextual return columns, parser-style structure, execution success, and LLM-judge review. In the full run, only 2 of 4,777 candidates were schema-invalid after deterministic governance, and all 3,000 exported examples passed the deterministic and judge gates.

Key artifacts. code: validator.py; code: cypher_parser.py; code: question_constraints.py; snapshot: failure taxonomy; +1 more

Status. Supported by code, tests, and full-run failure taxonomy.

Risk. Semantic correctness still depends on execution evidence and judge review; the completed 80-row audit is a calibration sample rather than an exhaustive proof.

- Claim 3.** Reverse grounding and structured workload seeds are designed to improve reliability under local-model constraints.

Evidence. Three FinBench/SNB ablation suites are complete and evidence-ready: the original target-50 suite, the corrected target-100 suite, and a seed-17 target-50 repeat. Each suite completed 14/14 graph/variant cells, reached every category target for all non-empty/non-unconstrained runs, and passed paper-readiness audits with logs, model IDs,

Profile	Added constraint	Intended evidence
Schema-only	Use only visible schema.	Baseline for schema-grounded prompting.
Instructions	Exact values, datatype rules, no nested aggregations.	Targets MTQ-style prompt refinements.
Examples	Placholderized retrieved NL-Cypher pairs.	Tests whether examples help without extra governance.
Examples + instructions	Few-shot plus explicit rules.	Closest controlled analogue to MTQ Table 5.
Full governed	Production-derived Cypher hints, AST-safe rewrites, judge gate.	PIPE-Cypher production setting.

Table 14: Prompt-profile plan inspired by Mind the Query’s prompt-variant analysis. Only completed, audited target-50-or-larger results should be promoted into results tables.

Dimension	Mind the Query	PIPE-Cypher
Generation review gate	Manual logical review	Deterministic gates + local LLM judge
Human effort	Reported 1,400 person-hours	80-row post-hoc judge calibration audit
Private values	Public benchmark values	Configurable sampling and redacted export
Refresh	Static dataset release	Rerunnable private benchmark factory
Model endpoint	Gemini for generation	Local Qwen3.5-9B endpoint

Table 15: Industry deployment contrast with Mind the Query. PIPE-Cypher focuses on private refreshable benchmark generation rather than a one-time public dataset.

code revisions, run summaries, collection manifests, and gate-rate availability. The target-100 suite is the detailed appendix ablation table/figure, while the three-suite comparison reports target-normalized coverage, acceptance variation, execution rates, and judge rates for stability.

Key artifacts. script: run live ablation suite; script: monitor remote ablation suite; script: monitor remote ablation queue; script: collect remote ablation suite; +20 more

Status. Supported by audited target-100 evidence and three-suite target-size/repeated-seed comparison.

Risk. The unconstrained local generation baseline produced no accepted examples under the strict executable gates, so comparisons against it should be framed as a gate-stress baseline rather than a tuned generation system.

- Claim 4.** The benchmark controls category and difficulty balance while exposing residual diversity limits.

Evidence. The full export has perfect category balance, planned 2:1 FinBench/SNB graph mix, and a near-even easy/medium split. Diversity diagnostics report Distinct-n, sampled self-BLEU-2, query-signature diversity, normalized entropy, schema coverage, structural feature rates, and aggregate value-grounding diagnostics. The full export uses 1,115 unique grounded entity values and exactly quotes grounded values in 82.6% of examples with entity bindings, making template and value concentration visible instead of hidden.

Setting	Graph	Records	Accepted	Acceptance	Categories at target
Unconstrained LLM	FinBench	0	0	0.000	0/8
Unconstrained LLM	SNB	0	0	0.000	0/8
Reverse-only	FinBench	815	800	0.982	8/8
Reverse-only	SNB	820	800	0.976	8/8
Validators+repair	FinBench	833	800	0.960	8/8
Validators+repair	SNB	824	800	0.971	8/8
No retrieval	FinBench	819	800	0.977	8/8
No retrieval	SNB	809	800	0.989	8/8
No rewrite	FinBench	826	800	0.969	8/8
No rewrite	SNB	834	800	0.959	8/8
No LLM judge	FinBench	812	800	0.985	8/8
No LLM judge	SNB	828	800	0.966	8/8
Full PIPE-Cypher	FinBench	824	800	0.971	8/8
Full PIPE-Cypher	SNB	824	800	0.971	8/8

Table 16: Live target-100 ablation evidence with local Qwen3.5-9B. Each graph run targets 100 accepted examples per category.

Setting	Graph	Read-only	Syntax	Schema	Exec.	Judge
Unconstrained LLM	FinBench	0.000	0.000	0.000	0.000	0.000
Unconstrained LLM	SNB	0.000	0.000	0.000	0.000	0.000
Reverse-only	FinBench	1.000	1.000	0.982	0.982	0.982
Reverse-only	SNB	1.000	1.000	0.998	0.998	0.976
Validators+repair	FinBench	1.000	1.000	1.000	1.000	0.960
Validators+repair	SNB	1.000	1.000	0.998	0.998	0.971
No retrieval	FinBench	1.000	1.000	1.000	1.000	0.977
No retrieval	SNB	1.000	1.000	0.998	0.998	0.989
No rewrite	FinBench	1.000	1.000	1.000	1.000	0.969
No rewrite	SNB	1.000	1.000	0.998	0.998	0.959
No LLM judge	FinBench	1.000	1.000	1.000	1.000	0.985
No LLM judge	SNB	1.000	1.000	0.998	0.998	0.966
Full PIPE-Cypher	FinBench	1.000	1.000	1.000	1.000	0.971
Full PIPE-Cypher	SNB	1.000	1.000	0.998	0.998	0.971

Table 17: Quality-gate rates for the live target-100 ablation suite. Rates are computed over all generated records in each graph/setting.

Key artifacts. snapshot: diversity report; paper table: diversity; paper figure: diversity diagnostics; paper figure: full export distribution

Status. Supported by full-export statistics and diversity diagnostics.

Risk. Low query-signature diversity in the reported local-model run remains a limitation and ablation target.

- Claim 5.** The generated benchmark is useful for downstream Text2Cypher evaluation.

Evidence. A zero-shot local Qwen3.5-9B downstream evaluation over the 296-example full test split reports parse validity, schema validity, execution success, execution accuracy, answer F1, per-category breakdowns, fixed-seed bootstrap uncertainty intervals, and a row-level error taxonomy. The model reaches 0.189 execution accuracy with a 95% bootstrap interval of [0.145, 0.233] and 0.622 execution success with interval [0.564, 0.676]. The error taxonomy classifies 240 incorrect rows into answer mismatches, execution failures, schema invalid predictions, and parse-invalid predictions, showing the benchmark can discriminate easy aggregation from harder operational categories and can explain model

Setting	Graph	Suites	Target cov.	Acceptance	Cat. target	Exec.	Judge
No LLM judge	FinBench	3/3	1.000	0.994±0.008	1.000	1.000	0.994
No retrieval	FinBench	3/3	1.000	0.967±0.031	1.000	1.000	0.967
No rewrite	FinBench	3/3	1.000	0.969±0.023	1.000	1.000	0.969
Full PIPE-Cypher	FinBench	3/3	1.000	0.975±0.016	1.000	1.000	0.975
Reverse-only	FinBench	3/3	1.000	0.994±0.011	1.000	0.994	0.994
Unconstrained LLM	FinBench	3/3	0.000	0.000	0.000	0.000	0.000
Validators+repair	FinBench	3/3	1.000	0.987±0.023	1.000	1.000	0.987
No LLM judge	SNB	3/3	1.000	0.982±0.015	1.000	0.996	0.982
No retrieval	SNB	3/3	1.000	0.993±0.004	1.000	0.996	0.993
No rewrite	SNB	3/3	1.000	0.983±0.021	1.000	0.996	0.983
Full PIPE-Cypher	SNB	3/3	1.000	0.987±0.014	1.000	0.996	0.987
Reverse-only	SNB	3/3	1.000	0.989±0.011	1.000	0.996	0.989
Unconstrained LLM	SNB	3/3	0.000	0.000	0.000	0.000	0.000
Validators+repair	SNB	3/3	1.000	0.987±0.014	1.000	0.996	0.987

Table 18: Target-size and repeated-seed ablation sensitivity. Target coverage normalizes accepted examples by each suite’s planned graph/category target, so target-50 and target-100 suites can be compared without treating larger raw counts as quality gains.

Audit packet property	Value
Rows	80
Judge accept / reject	40 / 40
FinBench / SNB rows	48 / 32
Easy / medium rows	26 / 54
Labeled rows	80
Agreement / Cohen’s κ	0.800 / 0.600
Judge precision / recall	1.000 / 0.714
Judge specificity	1.000
False accepts / false rejects	0 / 16

Table 19: Post-hoc judge calibration from the completed 80-row human audit. Labels are used only to calibrate the automated judge, not as a generation gate.

Metric	Point	95% CI	SE	N
Parse valid	0.959	[0.936, 0.980]	0.012	296
Schema valid	0.905	[0.872, 0.939]	0.017	296
Execution success	0.622	[0.564, 0.676]	0.028	296
Execution accuracy	0.189	[0.145, 0.233]	0.022	296
Answer F1	0.189	[0.149, 0.236]	0.023	296

Table 20: Bootstrap uncertainty for downstream Text2Cypher evaluation on the full exported test split. Intervals use row-level resampling with a fixed seed and are intended for appendix reporting.

Downstream outcome	Count	Share of incorrect
Answer mismatch	128	0.533
Execution failed	72	0.300
Schema invalid	28	0.117
Parse invalid	12	0.050

Table 21: Downstream Text2Cypher failure taxonomy for local Qwen3.5-9B on the full exported test split. Shares exclude exact-answer matches.

failure modes.

Key artifacts. downstream summary; snapshot: downstream uncertainty; snapshot: downstream error report; research note: downstream evaluation; +6 more

Status. Supported by live downstream evaluation against FinBench/SNB databases, with bootstrap uncertainty computed from row-level outputs.

Risk. Only one downstream model has been evaluated so far.

6. **Claim 6.** Automated LLM-judge review can replace human-in-the-loop generation gates while still exposing judge risk.

Evidence. The pipeline uses deterministic validation plus LLM-judge review as the generation gate. A post-hoc 80-row judge audit packet preserves 40 judge accepts, 40 judge

rejects, both graphs, and all eight categories. The completed human audit reports 80.0% agreement, Cohen’s kappa 0.60, balanced accuracy 0.857, judge precision 1.00, judge specificity 1.00, judge recall 0.714, zero false accepts, and 16 false rejects. The analyzer requires complete labels for paper-facing calibration, and the tracked summary avoids committing raw value-bearing audit rows.

Key artifacts. code: judge.py; code: calibration.py; code: judge_audit_packet.py; script: create judge annotation sheets; +6 more

Status. Supported by completed human audit summary.

Metric	PIPE-Cypher support	Primary citation
ROUGE-1/2/L	Reference overlap over serialized answer sets and query strings	(Lin, 2004)
BLEU	Sentence-level reference overlap with brevity penalty	(Papineni et al., 2002)
METEOR	Exact-token METEOR-style precision/recall with fragmentation penalty	(Banerjee and Lavie, 2005)
BERTScore	Optional contextual-embedding similarity when installed	(Zhang et al., 2020)
FrugalScore	Optional efficient learned NLG metric when installed	(Kamal Eddine et al., 2022)
Cosine	Token-count vector cosine similarity	(Manning et al., 2008)
Jaro-Winkler	Character-level string similarity for near-match diagnostics	(Jaro, 1989; Winkler, 1990)
Exact match	Raw and normalized string exact match	(Rajpurkar et al., 2016)

Table 22: Supplementary reference-based text metrics supported by the PIPE-Cypher evaluator. These metrics are useful for debugging answer rendering, paraphrase sensitivity, and near-match behavior, but they do not replace execution accuracy or answer-set F1 for Text2Cypher correctness. BERTScore and FrugalScore are optional integrations because they require additional metric/model packages.

Risk. The judge appears conservative on this sample; larger audits or different enterprise schemas may reveal false accepts.

7. **Claim 7.** PIPE-Cypher supports arbitrary enterprise schema onboarding with configurable privacy redaction and value-sampling policies.

Evidence. The repo includes an enterprise deployment guide, an enterprise template config, privacy config fields, schema introspection that reads categorical-value thresholds, maximum sampled value length, and omitted-property patterns from config, deterministic redaction helpers, and a redacted benchmark export CLI. The redaction policy replaces quoted Cypher literals, question mentions, entity values, and string-valued result samples with stable placeholders while preserving query structure and gate metadata. The repo now also includes an ICIJ Offshore Leaks onboarding profile, config, download/load helpers, graph plan, schema-derived sparse-category templates, and a completed sanitized target-100 onboarding audit on a public finance/compliance graph beyond the LDBC study workloads.

Key artifacts. research note: enterprise deployment guide; research note: icij offshore-leaks graph plan; enterprise_template.yaml; schema_icij_offshoreleaks.json; +21 more

Status. Supported by docs, config, code, deterministic tests, a concrete third-graph onboarding plan, and a completed sanitized ICIJ target-100 live onboarding run with 800/983 accepted examples and 8/8 categories at target.

Risk. ICIJ is public rather than pri-

vate/anonymized enterprise data. The strongest industry validation would still come from a real tenant graph under the same audit protocol.

C Prompt Contracts

PIPE-Cypher treats prompts as versioned implementation artifacts. The list summarizes the prompt contracts used for generation, repair, judging, and downstream evaluation; hashes fingerprint the full prompt constants in the codebase.

1. **Template generation.** Stage: Workload proposal. SHA-256: 10b25b.

Contract. Schema-only labels, relationships, properties, and categorical values; Realistic enterprise analyst wording; At most two typed slots and JSON-only output

2. **Reverse binding.** Stage: Graph grounding. SHA-256: 48e949.

Contract. Read-only MATCH/WHERE/RETURN DISTINCT/LIMIT only; Slot variables named exactly as requested; Forward relationship directions from the schema

3. **Cypher generation.** Stage: Candidate query. SHA-256: 61c557.

Contract. Only schema-visible constructs and observed directions; RETURN DISTINCT for set returns and exact equality for quoted values; Context columns, categorical hints, placeholderized retrieval, and no writes

4. **Repair.** Stage: Validation feedback. SHA-256: 56c419.

Contract. Preserve question intent while fixing validation or execution issues; Keep query

685	read-only and schema-grounded; Return only	Structure: join_heavy, aggregation.	735
686	corrected Cypher	Relationships: OWN_ACCOUNT, TRANS-	736
687		FER_TO.	737
688	5. LLM judge . Stage: Quality gate. SHA-256:	Gates: RO/Syn/Schema/Exec/Judge.	738
689	421c7b.	Result sample: {TotalTransferredAmount:	739
690	Contract. Inputs include question, Cypher,	14972318.41}; observed rows: 1	740
691	schema slice, execution rows, and validation		
692	summary; Strict JSON scores for ambiguity,	3. FinBench / Complex Retrieval / easy. Ques-	741
693	semantic alignment, schema use, and diffi-	tion: Which accounts received transfers from	742
694	culty; Categorical values constrain query liter-	accounts owned by person 'Zof'?	743
695	als, not observed result-row values; Pass only		
696	useful, unambiguous enterprise benchmark	MATCH (p:Person {personName:	744
697	examples	'Zof'})	745
698		-[:OWN_ACCOUNT]-> (src:Account)	746
699		-[:TRANSFER_TO]-> (dst:Account)	747
700	6. Downstream Text2Cypher . Stage: Model	RETURN DISTINCT dst.accountId AS	748
701	evaluation. SHA-256: 4c07ff.	AccountId, dst.accountType AS	749
702	Contract. Read-only Cypher only; Schema-	AccountType, dst.isBlocked AS	750
703	visible constructs and exact direction preser-	IsBlocked	751
704	vation; RETURN DISTINCT, count/ranking	LIMIT 300	752
705	rules, and no explanations	Structure: join_heavy, bounded_result.	753
706		Relationships: OWN_ACCOUNT, TRANS-	754
707		FER_TO.	755
708		Gates: RO/Syn/Schema/Exec/Judge.	756
709		Result sample: { AccountId:	757
710		4687402787162554854, AccountType:	758
711		credit card, IsBlocked: False}; observed rows:	759
712		3	760
713	1. FinBench / Boolean Existence / easy. Ques-	4. FinBench / Negation Difference / medium.	761
714	tion: Does account '187743809466009406'	Question: Which accounts owned by person	762
715	have any outgoing transfer?	'Kant' have not sent any transfers?	763
716			
717	MATCH (src:Account {accountId:	MATCH (p:Person {personName:	764
718	'187743809466009406'})	'Kant'})	765
719	-[:TRANSFER_TO]->	-[:OWN_ACCOUNT]-> (a:Account)	766
720	(:Account)	WHERE NOT (a) -[:TRANSFER_TO]->	767
721	RETURN DISTINCT COUNT(DISTINCT src)	(:Account)	768
722	> 0	RETURN DISTINCT a.accountId AS	769
723	AS HasOutgoingTransfer	AccountId, a.accountType AS	770
724		AccountType,	771
725		a.isBlocked AS IsBlocked	772
726		LIMIT 300	773
727		Structure: join_heavy, negation,	774
728		bounded_result.	775
729		Relationships: OWN_ACCOUNT, TRANS-	776
730		FER_TO.	777
731		Gates: RO/Syn/Schema/Exec/Judge.	778
732		Result sample: { AccountId:	779
733		4739757132830738341, AccountType:	780
734		merchant account, IsBlocked: False};	781
		observed rows: 1	782
		5. FinBench / Path Temporal / medium. Ques-	783
		tion: Which accounts can receive money	784
		within two transfer hops from accounts owned	785
		by person 'Sossamon'?	786

787	MATCH (p:Person {personName:	844	8. FinBench / Simple Retrieval / easy. Question: Which accounts are owned by person
788	'Sossamon'}) -[:OWN_ACCOUNT]->	845	'Barry'?
789	(src:Account) -[:TRANSFER_TO*1..2]->	846	
790	(dst:Account)		
791	RETURN DISTINCT dst.accountId AS	847	
792	AccountId, dst.accountType AS	848	
793	AccountType, dst.isBlocked AS	849	
794	IsBlocked	850	
795	LIMIT 300	851	
		852	
796	Structure: join_heavy, path, bounded_result.	853	
797	Relationships: OWN_ACCOUNT, TRANS-	854	
798	FER_TO.		
799	Gates: RO/Syn/Schema/Exec/Judge.		
800	Result sample: {AccountId:	855	Structure: single_hop, bounded_result.
801	4687402787162554854, AccountType:	856	Relationships: OWN_ACCOUNT.
802	credit card, IsBlocked: False}; observed rows:	857	Gates: RO/Syn/Schema/Exec/Judge.
803	4	858	Result sample: {AccountId:
804		859	4732438783436260772, AccountType:
805		860	internet account, IsBlocked: False}; observed
806		861	rows: 1
807		862	9. SNB / Boolean Existence / medium. Question: Does person with id 6597069766828
808	MATCH (p:Person {personName:	863	like any post?
809	'Barry'})	864	
810	-[:OWN_ACCOUNT]->	865	
811	(src:Account) -[:TRANSFER_TO]->	866	
812	(:Account)	867	
813	WITH src, SUM(t.amount) AS	868	
814	totalAmount	869	
815	RETURN DISTINCT src.accountId AS	870	
816	AccountId, src.accountType AS		
817	AccountType, src.isBlocked AS		
818	IsBlocked,		
819	totalAmount		
820	ORDER BY totalAmount DESC		
821	LIMIT 1		
822	Structure: join_heavy, aggregation, or-	871	Structure: single_hop, aggregation, optional.
823	der_rank, bounded_result.	872	Relationships: LIKES.
824	Relationships: OWN_ACCOUNT, TRANS-	873	Gates: RO/Syn/Schema/Exec/Judge.
825	FER_TO.	874	Result sample: {LikesAnyPost: True}; ob-
826	Gates: RO/Syn/Schema/Exec/Judge.	875	erved rows: 1
827	Result sample: {AccountId:	876	10. SNB / Complex Aggregation / medium.
828	4732438783436260772, AccountType:	877	Question: How many distinct posts
829	internet account, IsBlocked: False}; observed	878	are in forums joined by person with id
830	rows: 1	879	6597069766845?
831			
832	7. FinBench / Simple Aggregation / easy. Question: How many accounts are owned by person 'Kaewsuktae'?	880	
833		881	
834	MATCH (p:Person {personName:	882	
835	'Kaewsuktae'}) -[:OWN_ACCOUNT]->	883	
836	(a:Account)	884	
837	RETURN DISTINCT COUNT(DISTINCT a) AS	885	
838	AccountCount	886	
839	Structure: single_hop, aggregation.	887	Structure: join_heavy, aggregation.
840	Relationships: OWN_ACCOUNT.	888	Relationships: CONTAINER_OF,
841	Gates: RO/Syn/Schema/Exec/Judge.	889	HAS_MEMBER.
842	Result sample: {AccountCount: 1}; observed	890	Gates: RO/Syn/Schema/Exec/Judge.
843	rows: 1	891	Result sample: {JoinedForumPostCount: 1};
		892	observed rows: 1
		893	11. SNB / Complex Retrieval / easy. Question:
		894	Which people are members of forums contain-
		895	ing posts tagged 'Manuel_Noriega'?

896	MATCH (forum:Forum) -[:HAS_MEMBER]->	WITH forum, COUNT(DISTINCT person)	951
897	(p:Person), (forum)	AS	952
898	-[:CONTAINER_OF]->	memberCount	953
899	(post:Post) -[:HAS_TAG]-> (tag:Tag	RETURN DISTINCT forum.title AS	954
900	{name: 'Manuel_Noriega'})	ForumTitle, memberCount	955
901	RETURN DISTINCT p.id AS PersonId	ORDER BY memberCount DESC	956
902	LIMIT 200	LIMIT 1	957
903	<i>Structure:</i> join_heavy, bounded_result.	<i>Structure:</i> join_heavy, aggregation, or-	958
904	<i>Relationships:</i> CONTAINER_OF,	der_rank, bounded_result.	959
905	HAS_MEMBER, HAS_TAG.	<i>Relationships:</i> HAS_MEMBER, HAS_TAG.	960
906	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	961
907	<i>Result sample:</i> {PersonId: 4398046511220};	<i>Result sample:</i> {ForumTitle: Wall of Javed	962
908	observed rows: 19	Khan, memberCount: 33}; observed rows: 1	963
909	12. SNB / Negation Difference / medium. Ques-	15. SNB / Simple Aggregation / easy. Question:	964
910	<i>tion:</i> Which members of forum 'Wall of Celso	How many posts are tagged 'Vietnam'?	965
911	Oliveira' have not liked any post?		
912	MATCH (forum:Forum {title: 'Wall of	MATCH (post:Post) -[:HAS_TAG]->	966
913	Celso Oliveira'}) -[:HAS_MEMBER]->	(tag:Tag	967
914	(p:Person)	{name: 'Vietnam'})	968
915	WHERE NOT (p) -[:LIKES]-> (:Post)	RETURN DISTINCT COUNT(DISTINCT post)	969
916	RETURN DISTINCT p.id AS PersonId,	AS	970
917	p.firstName AS FirstName, p.lastName	PostCount	971
918	AS		
919	LastName	<i>Structure:</i> single_hop, aggregation.	972
920	LIMIT 200	<i>Relationships:</i> HAS_TAG.	973
921	<i>Structure:</i> join_heavy, negation,	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	974
922	bounded_result.	<i>Result sample:</i> {PostCount: 1}; observed	975
923	<i>Relationships:</i> HAS_MEMBER, LIKES.	rows: 1	976
924	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	16. SNB / Simple Retrieval / easy. Ques-	977
925	<i>Result sample:</i> {FirstName: K., LastName:	<i>tion:</i> Which post IDs did person with id	978
926	Sen, PersonId: 94}; observed rows: 1	4398046511124 like?	979
927	13. SNB / Path Temporal / easy. Question:	MATCH (p:Person {id:	980
928	Which people are within two knows hops of	4398046511124})	981
929	person with id 4398046511136?	-[:LIKES]-> (post:Post)	982
930	MATCH (src:Person {id:	RETURN DISTINCT post.id AS PostId	983
931	4398046511136})	LIMIT 200	984
932	-[:KNOWS*1..2]-> (dst:Person)		
933	RETURN DISTINCT dst.id AS PersonId,	<i>Structure:</i> single_hop, bounded_result.	985
934	dst.firstName AS FirstName,	<i>Relationships:</i> LIKES.	986
935	dst.lastName	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	987
936	AS LastName	<i>Result sample:</i> {PostId: 343597385744}; ob-	988
937	LIMIT 200	served rows: 5	989
938	<i>Structure:</i> single_hop, path, bounded_result.		
939	<i>Relationships:</i> KNOWS.		
940	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.		
941	<i>Result sample:</i> {FirstName: Rafael,		
942	LastName: Fernández, PersonId:		
943	4398046511333}; observed rows: 25		
944	14. SNB / Ranking Topk / medium. Ques-		
945	<i>tion:</i> Among forums tagged 'James_K._Polk',		
946	which forum has the most members?		
947	MATCH (forum:Forum) -[:HAS_TAG]->		
948	(tag:Tag {name: 'James_K._Polk'})		
949	MATCH (forum) -[:HAS_MEMBER]->		
950	(person:Person)		

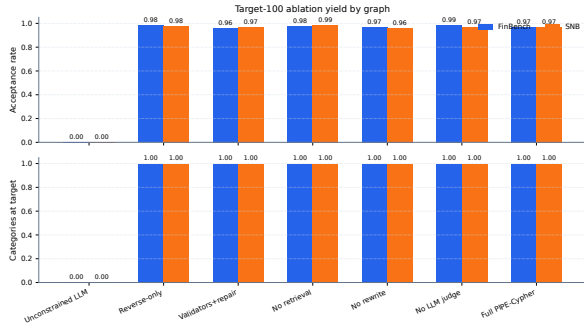


Figure 2: Audited target-100 ablation suite over FinBench and SNB. Every non-unconstrained graph/setting cell reaches all eight category targets with 800 accepted examples per cell.

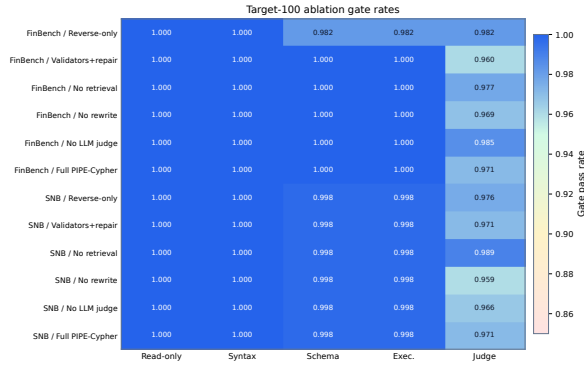


Figure 3: Gate-rate heatmap for the audited target-100 ablation suite. Deterministic read-only and syntax gates are saturated; execution and judge rates expose the remaining quality differences across graph and pipeline variants.

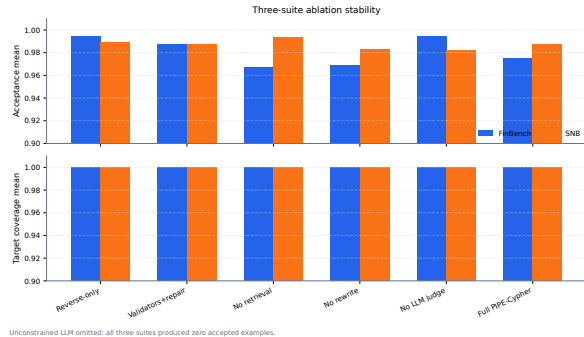


Figure 4: Three-suite ablation stability over the original target-50 suite, corrected target-100 suite, and seed-17 target-50 repeat. Target-normalized coverage stays at 1.000 for all non-unconstrained cells.

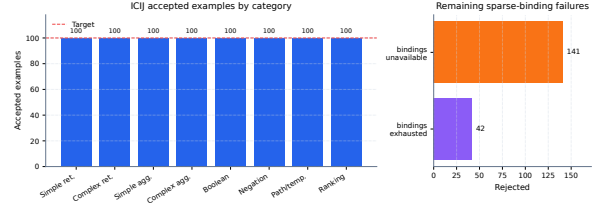


Figure 5: ICIJ Offshore Leaks onboarding audit. Schema-derived sparse-category templates recover balanced target-100 coverage on a public finance/compliance graph beyond the two LDBC workloads.

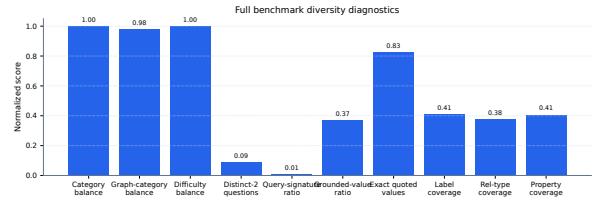


Figure 6: Diversity diagnostics for the full 3,000-example export. Category, graph-category, and difficulty balance are strong by construction; schema coverage and query-signature diversity expose remaining concentration from the seeded-template run.

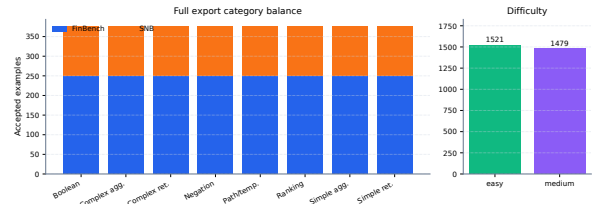


Figure 7: Full 3,000-example export distribution. The benchmark preserves the planned 2:1 FinBench/SNB graph mix while balancing all eight Cypher workload categories and maintaining a near-even easy/medium difficulty split.

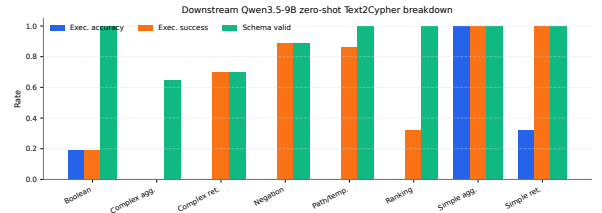


Figure 8: Downstream zero-shot Qwen3.5-9B Text2Cypher evaluation by workload category on the full test split. The breakdown separates final execution accuracy from parse/schema/execution success signals, making difficult operational categories visible.

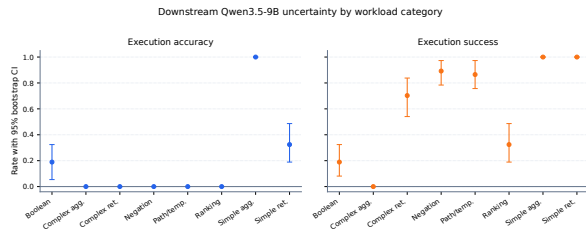


Figure 9: Bootstrap uncertainty for downstream Qwen3.5-9B Text2Cypher evaluation by workload category. Error bars show 95% row-level bootstrap intervals over the full exported test split.

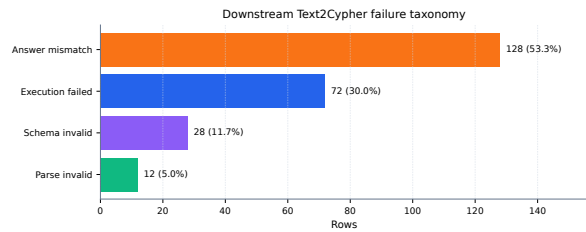


Figure 10: Downstream Text2Cypher failure taxonomy for the local Qwen3.5-9B baseline on the full test split. Exact matches are excluded so the chart exposes whether misses come from invalid Cypher, failed execution, or semantically wrong executable queries.

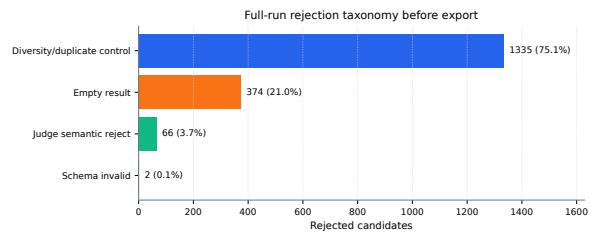


Figure 11: Full-run rejection taxonomy before benchmark export. Most rejected candidates come from duplicate/diversity control during category recovery and from empty execution results; schema-invalid Cypher is rare after deterministic governance.